Adithya T S

Software Engineer | Generative AI Engineer

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tsadithya24

PROFESSIONAL SUMMARY

Results-driven Software Engineer with 3.5+ years of experience designing high-performance backend, computer vision, and AI-powered systems across Windows and Linux environments. Experienced in building scalable AI architectures involving multi-agent orchestration, Retrieval-Augmented Generation (RAG), OCR pipelines, and autonomous testing frameworks using Python and modern LLM ecosystems. Hands-on expertise in LangChain, FAISS, Hugging Face Transformers, multimodal AI workflows, and enterprise-grade system integration. Strong background in distributed systems, multithreading, API integration, and cloud-connected applications, with growing specialization in agentic AI systems, tool orchestration, and LLM-driven automation

SKILLS

- **Languages:** C++, Python, SQL
- **Computer Vision & Imaging:** OpenCV, Image Processing, Video Streaming, OCR Pipelines, Camera SDK Development
- **AI & Deep Learning:** Large Language Models (LLMs), Multi-Agent AI Systems, Agent Orchestration, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Hugging Face Transformers, FAISS, Ollama, LlamaIndex, Prompt Engineering, Tool Calling, Function Calling, Multimodal AI Pipelines
- **Frameworks & Multimedia:** DirectShow, Windows Media Foundation, OpenGL
- **Distributed Systems:** Multithreading, Parallel Computing, Synchronization, Fault Tolerance
- **Databases:** MySQL, SQLite
- **Testing & CI/CD:** Google Test, Jenkins, GitLab CI
- **Tools & Platforms:** Git, GitHub, GitLab, Windows, Linux
- **Methodologies:** Agile Scrum, JIRA

WORK EXPERIENCE

E-CON SYSTEMS INDIA PVT LTD

Project Engineer — Dec 2022 - Present

- Developed and optimized high-performance imaging pipelines and camera SDK modules using C++ and multithreading, improving backend throughput by 35% and reducing integration time by 30%.
- Built real-time image processing and visualization applications for industrial camera systems using OpenCV, DirectShow, and Windows Media Foundation.
- Worked on AI-assisted imaging and automation workflows integrating computer vision pipelines with backend orchestration systems.
- Collaborated on scalable system design involving real-time processing, resource optimization, and high-throughput data pipelines.
- Contributed to automation tooling and intelligent monitoring systems for production-grade imaging applications.
- Worked extensively with camera streaming, frame synchronization, pixel format handling, and real-time rendering pipelines under Windows environments.
- Designed and implemented AWS S3 upload services using C++ and libcurl with secure SSL/TLS validation and region-aware operations.
- Built robust microservices for storage management and device status monitoring, enabling automated cleanup workflows and real-time hardware state signaling.
- Engineered a Python-based Cloud Image Viewer with intelligent caching and live refresh support for scalable remote image monitoring.
- Contributed to CI/CD pipelines, unit testing, and automated validation workflows using Jenkins, Google Test, and GitLab CI, achieving 90%+ unit test coverage.
- Collaborated in Agile teams to debug and optimize real-world imaging issues across varying hardware and deployment conditions.

PROJECTS

TestAny - AI Powered Autonomous GUI Testing Framework Jan 2026 - Mar 2026

- Architected a multi-agent AI system consisting of Planner, Vision, Executor, and Validation agents for autonomous desktop application testing workflows.
- Designed agent orchestration pipelines enabling task planning, execution, UI understanding, and automated validation using LLM-driven coordination strategies.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using FAISS, LangChain, and Hugging Face Transformers for context-aware test generation from product documentation and historical test artifacts.
- Developed multimodal AI pipelines integrating OCR, computer vision, and reasoning workflows using PaddleOCR, EasyOCR, Tesseract, and OpenCV.
- Worked on tool-based orchestration and agent communication patterns for automated execution and validation loops.
- Built resilient fallback-based vision and reasoning pipelines to improve robustness under noisy real-world UI conditions.
- Explored context-window optimization, prompt engineering, and memory-aware workflows for long-running autonomous testing tasks.

Stack: Python, LangChain, FAISS, Hugging Face Transformers, PaddleOCR, OpenCV, PyAutoGUI

AWARD

Extra Mile Award E-CON SYSTEMS INDIA PVT LTD

Recognized for developing a production-ready Windows motion detection application ahead of schedule, delivering an enhanced feature set that improved customer satisfaction and reduced manual monitoring requirements.

EDUCATION

SRM University

Master of Technology (Data Science) - 9.5 Sept 2024 - May 2026

Chennai Institute of Technology

Bachelor of Engineering (Computer Science and Engineering) - 8.48 Jun 2018 - July 2022

LANGUAGES KNOWN

English , Hindi , Tamil